Pre-A Q3LOCK Fixed Standard Form, C0 Completion, and Homological Generator Boundary Checkpoint

TECT verification-first repository
claim `C6-SPACETIME-SIGNATURE`

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1 Purpose and scope

This R-167-only gate-level checkpoint synthesizes `EXP-000818` and advances reusable result `R-167 v2.4` without changing a claim card or tier. The result is additive, `T0`, and `claim_bearing: false`. Its stable result identifier is

> `PA-CP1-ST8-Q3LOCK-SECOND-WEIGHTED-ENERGY-MOMENT-AND-COMMON-ALPHA-CAUCHY-GATE-SPLIT`.

R-167 v2.4 preserves every v2.3 and earlier theorem, route boundary, negative result, and issued source/PDF pair. It closes exactly three additional scoped children. First, the two ordinary Hilbert–Schmidt implementer legs from v2.3 give strong-star convergence in each fixed finite faithful Gibbs standard representation and point-strong-star convergence of conjugation by every fixed bounded observable. Second, a deterministic conditional theorem turns actual bidirectional all-shape point-norm Cauchy control on one common C-star algebra into a unique exhaustion-independent C_0 automorphism group. Third, the first parity-equivariant rank-two homological generator is edge local, has an explicit QPS norm and Ritz-tail bound, and reproduces the v2.3 onsite-resolvent coefficient in its second-order low block.

Two exact fixtures separately reject automatic all-order promotions. A disconnected retained low-block spectator creates a mixed coefficient in the raw global scalar Feshbach map, and finite harmonic Ritz boundedness fails to give cutoff-uniform ordinary operator-norm Schrieffer–Wolff smallness. These are narrow implication failures, not no-go theorems for linked-cluster, relative-form, graph-norm, or resolvent-compatible Q3 constructions.

The contemporaneous `EXP-000817` record is an authority-link correction only. It adds no theorem, reusable-result version, negative result, claim-card change, or tier change. In particular, the four historical `EXP-000790` scoped children remain phasewise or finite-regulator statements and do not imply a common alpha. `EXP-000818`, not `EXP-000817`, is the theorem-bearing exploration for this checkpoint.

R-168 v1.3 remains historical and is not reissued. No R-168 numerical total, raw script hash, new theorem, or lifecycle metadata is part of this document.

2 Closed children, negative authorities, and surviving parents

The three children closed by `EXP-000818` are exactly

1. `PA-CP1-ST8-Q3LOCK-FIXED-FINITE-FAITHFUL-GIBBS-STANDARD-FORM-POINT-STRONGSTAR-OBSERVABLE-CUTOFF-REMOVAL`;
2. `PA-CP1-ST8-Q3LOCK-CONDITIONAL-BIDIRECTIONAL-ALL-SHAPE-POINT-NORM-CAUCHY-CO-AUTOMORPHISM-COMPLETION`;

3. PA-CP1-ST8-Q3LOCK-FIRST-LOCAL-HOMOLOGICAL-RANK-TWO-GENERATOR-QPS-NORM-AND-RITZ-CUTOFF.

The two new scoped negative authorities are exactly

1. NG-2026-08-12-PRE-A-ST8-Q3LOCK-SECOND-ORDER-DISJOINT-VANISHING-AUTOMATIC-ALL-ORDER-GLOBAL-FESHBACH-CONNECTEDNESS;
2. NG-2026-08-12-PRE-A-ST8-Q3LOCK-RITZ-CUTOFF-ORDINARY-BOUNDED-OPERATOR-SW-SMALLNESS-UNIFORMITY.

The following load-bearing parents and successors remain **OPEN**:

1. PA-CP1-ST8-Q3LOCK-LOCAL-STRICT-ALL-EXHAUSTION-TWO-ORIENTATION-HISTORY-COMMON-ALPHA;
2. PA-CP1-ST8-Q3LOCK-CONNECTED-RANK-TWO-OSCILLATOR-ELIMINATION-QPS-NORM-AND-CUTOFF-COMPATIBILITY;
3. PA-CP1-ST8-Q3LOCK-INFINITE-DIMENSIONAL-RANK-TWO-BAND-BLOCK-DIAGONALIZATION-AND-TWO-PHASE-QPS;
4. PA-CP1-ST8-Q3LOCK-BROKEN-SECTOR-GNS-GAP-COERCIVITY;
5. PA-ROUND1-EVIDENCE-ROLE-AND-MINIMUM-MANIFEST-FREEZE.

The three closed children are reusable local results. Their conjunction does not close any parent in this list.

3 Fixed finite faithful Gibbs standard-form theorem

Fix one finite periodic Q3 volume, one compact source, one inverse temperature, one declared split mesh/history, and one radius R_0. Let ρ be its faithful Gibbs density on $\mathcal{H}$, let

$$\mathcal{K} = \mathfrak{S}_2(\mathcal{H}), \qquad \xi = \rho^{1/2}, \qquad \pi(A)X = AX, \tag{3.1}$$

and let P_L, P be the cutoff and exact split implementers. Define

$$e_+(L) = \|(P - P_L)\xi\|_2, \qquad e_-(L) = \|(P^* - P_L^*)\xi\|_2 = \|\xi(P - P_L)\|_2. \tag{3.2}$$

The equality in the second leg uses only invariance of the Hilbert–Schmidt norm under adjoint. It does not assume modular commutation. At fixed R_0, the v2.3 opposite telescopes give separately

$$e_\pm(L) \le \frac{54|c|T\sqrt{M_{20}}}{\hbar} R_0^3 L^{-8}, \tag{3.3}$$

and their paired bound is

$$e_+(L) + e_-(L) \le \frac{54\sqrt{2}|c|T\sqrt{M_{20}}}{\hbar} R_0^3 L^{-8}. \tag{3.4}$$

Faithfulness makes ξ cyclic for the right commutant. More explicitly,

$$\{\xi C : C \in B(\mathcal{H})\} \quad \text{is dense in } \mathfrak{S}_2(\mathcal{H}), \tag{3.5}$$

because finite-rank operators are dense and their left vectors can be approximated by vectors in the dense range of ξ. On this core,

$$\|(P - P_L)\xi C\|_2 \le e_+(L)\|C\|, \qquad \|(P^* - P_L^*)\xi C\|_2 \le e_-(L)\|C\|. \tag{3.6}$$

All implementers are unitary, so uniform boundedness extends both convergences to every vector of $\mathfrak{S}_2(\mathcal{H})$. Left multiplication by P_L therefore converges strong-star to left multiplication by P.

For fixed $A \in B(\mathcal{H})$ and fixed $\eta \in \mathfrak{S}_2(\mathcal{H})$,

$$\begin{aligned}&(P_L^* A P_L - P^* A P)\eta \\ &\quad = P_L^* A(P_L - P)\eta + (P_L^* - P^*)AP\eta \longrightarrow 0.\end{aligned} \tag{3.7}$$

Replacing A by A^* gives the adjoint statement. Hence

$$\boxed{\pi(P_L^* A P_L) \longrightarrow \pi(P^* A P) \quad \text{point-strong-star for every fixed } A.} \tag{3.8}$$

The quantitative claim stops at (3.3)–(3.6). Equation (3.8) is qualitative and is not an arbitrary-observable Gibbs-L2 channel estimate. Moreover, the optimized choice $L = R^{2/5}$ changes the volume, Gibbs density, and standard representation. Its v2.3 $R^{-1/5}$ implementer corridor is not a strong-star rate on one common Hilbert space and proves no thermodynamic common alpha.

For the exact fixed-member fixture $R_0 = 32$, the per-leg coefficients at $L = 4, 8$ are 27 and 27/256, while the paired coefficients are $27\sqrt{2}$ and $27\sqrt{2}/256$. An independent two-dimensional probe uses

$$\rho = \mathrm{diag}(1/5, 4/5), \qquad U_n = \frac{1}{n^2+1}\begin{pmatrix} n^2-1 & -2n \\ 2n & n^2-1 \end{pmatrix}. \tag{3.9}$$

For $P = I$, both squared legs are $4/(n^2+1)$, equal to 2/5 and 2/25 at $n = 3, 7$. With $A = \mathrm{diag}(1, -1)$, the squared weighted conjugation errors are $16n^2/(n^2+1)^2$, equal to 36/25 and 196/625. This finite probe checks the two directions and their decay; it does not prove the general theorem.

4 Conditional bidirectional all-shape C0 completion

Let $\mathcal{A}$ be one common unital C-star algebra, let $\mathcal{A}_0$ be a norm-dense unital star subalgebra, and let $\mathcal{D}$ be the directed set of every declared finite shape and cutoff. For each $\lambda \in \mathcal{D}$, let α_λ^t be a point-norm continuous automorphism group of $\mathcal{A}$. For every $A \in \mathcal{A}_0$ and finite T, assume that both nets

$$\alpha_\lambda^t(A), \qquad \alpha_\lambda^{-t}(A) \tag{4.1}$$

are point-norm Cauchy uniformly for $0 \le t \le T$ over the full directed set. This is stronger than convergence along one preferred exhaustion.

Every α_λ^t is isometric, so the two limits extend uniquely from $\mathcal{A}_0$ to $\mathcal{A}$; denote them by α^t and β^t. Norm limits preserve linearity, products, stars, and the unit. For the group law, add and subtract one fixed limiting argument:

$$\begin{aligned}&\|\alpha_\lambda^s(\alpha_\lambda^t(A)) - \alpha^s(\alpha^t(A))\| \\ &\quad \le \|\alpha_\lambda^t(A) - \alpha^t(A)\| + \|\alpha_\lambda^s(\alpha^t(A)) - \alpha^s(\alpha^t(A))\|.\end{aligned} \tag{4.2}$$

Both terms tend to zero. Passing the finite-volume identities through the norm limit gives

$$\alpha^{s+t} = \alpha^s\alpha^t, \qquad \alpha^t\beta^t = \beta^t\alpha^t = \mathrm{id}\,. \tag{4.3}$$

Thus $\beta^t = \alpha^{-t}$, every α^t is an automorphism, and the limit is exhaustion independent. Uniform convergence on compact time intervals preserves continuity on $\mathcal{A}_0$. A fixed $\mathcal{A}_0$ approximation and isometry leave a $2\|A - A_0\|$ error for a general A, so the limit is a point-norm continuous C_0 automorphism group.

For the arithmetic probe, take $H_n = (1 + 1/n)\,\mathrm{diag}(0, 1)$ on M_2. At $n = 10$, $m = 20$, and $T = 1$, $\|H_n - H_m\| = 1/20$; the Duhamel automorphism bound is $1/10$ for both time signs, and the bound from $n = 10$ to the limiting Hamiltonian is $1/5$.

This theorem is conditional. It proves no instance of (4.1) for Q3, no identification of the abstract generator with a local derivation or its domain, and no convergence or quotient theorem for finite-volume Gibbs/KMS states.

5 First local parity-equivariant homological generator

Let P be the product rank-two low-band projection, $Q = 1 - P$, and

$$KP = 0, \qquad K = \sum_x k_x \geq \Gamma Q, \qquad R = Q(QKQ)^{-1}Q. \tag{5.1}$$

For every edge e, define

$$T_e = QB_eP, \qquad \|T_e\| \leq \epsilon, \qquad D_e = RT_e, \qquad G_e = D_e - D_e^*. \tag{5.2}$$

Assume that K, P, and every B_e commute with the declared global parity. Then G_e is parity equivariant and skew adjoint. The global resolvent R is retained in every commutator. Locality follows because K preserves exact onsite high-support sectors and T_e creates high support only in e; on $\mathrm{ran}(T_e)$, the restriction RT_e is represented by the corresponding edge-local onsite resolvent.

Relative to $P \oplus Q$, G_e is one skew off-diagonal block matrix, so its norm is sharp at the block level:

$$\|G_e\| = \|D_e\| \leq \frac{\epsilon}{\Gamma}, \qquad \boxed{\|G\|_a \leq 2ze^a \frac{\epsilon}{\Gamma}}. \tag{5.3}$$

Let Π_M preserve parity, contain P, and commute with the onsite resolvent on the declared sectors. Put

$$G_{e,M} = R\Pi_M T_e - T_e^* \Pi_M R, \qquad \tau_M = \sup_e \|(1 - \Pi_M)T_e\|. \tag{5.4}$$

The same block identity gives

$$\|G_e - G_{e,M}\| \leq \frac{\tau_M}{\Gamma}, \qquad \|G - G_M\|_a \leq 2ze^a \frac{\tau_M}{\Gamma}. \tag{5.5}$$

Since $KD_e = T_e$, $D_eK = 0$, $KD_e^* = 0$, and $D_e^*K = T_e^*$, one obtains, for $G = \sum_e G_e$ and $V_{\mathrm{od}} = \sum_e (T_e + T_e^*)$,

$$[K, G] = V_{\mathrm{od}}, \qquad [G, K] = -V_{\mathrm{od}}. \tag{5.6}$$

The exact second-order low block after first-order cancellation is

$$\boxed{\frac{1}{2} P[G, V_{\mathrm{od}}]P = -\sum_{e,f} T_e^* R T_f,} \tag{5.7}$$

which is exactly the v2.3 onsite-resolvent coefficient.

For $z = 6$, $e^a = 2$, $\epsilon = 1/100$, $\Gamma = 2$, and $\tau_M = 1/1000$,

$$\|G_e\| \leq \frac{1}{200}, \quad \|G\|_a \leq \frac{3}{25}, \quad \|G_e - G_{e,M}\| \leq \frac{1}{2000}, \quad \|G - G_M\|_a \leq \frac{3}{250}. \tag{5.8}$$

On the four-level parity fixture

$$P = \operatorname{diag}(1,1,0,0), \quad K = \operatorname{diag}(0,0,2,2), \quad \Theta = \operatorname{diag}(1,-1,1,-1), \tag{5.9}$$

take $T = \epsilon(|3\rangle\langle 1| + |4\rangle\langle 2|)$. Direct matrix multiplication verifies parity, skew adjointness, (5.6), and

$$\frac{1}{2}P[G, T+T^*]P = -\frac{\epsilon^2}{\Gamma}P = -\frac{1}{20000}P. \tag{5.10}$$

Equation (5.7) is an algebraic coefficient identity. It proves no third-order coefficient, convergent BCH or Lie–Schwinger series, all-order connected interaction, phasewise intertwiner, or oscillator GNS gap.

6 Two exact all-order promotion boundaries

6.1 Disconnected spectator in a scalar Feshbach denominator

Let subsystem X have two low and two high states, high energy $\Gamma = 2$, and low-to-high coupling

$$T_X = \epsilon \operatorname{diag}(1,2), \qquad \epsilon = \frac{1}{10}. \tag{6.1}$$

Let the disconnected spectator Y lie entirely in the retained low block, with $h_Y = \operatorname{diag}(0,1)$. The original Hamiltonian is an X term plus a Y term and has no mixed interaction. With the global product low projection and scalar energy $E = 0$, the raw Feshbach self-energy is

$$\Sigma(0) = \epsilon^2 \operatorname{diag}(1,4) \otimes \operatorname{diag}(1/2, 1/3). \tag{6.2}$$

Using $Z = \operatorname{diag}(1,-1)$, its mixed $Z_X \otimes Z_Y$ coefficient is

$$\frac{\epsilon^2}{4}(1-4)(1/2-1/3) = -\frac{\epsilon^2}{8} = -\frac{1}{800}. \tag{6.3}$$

Thus the raw Feshbach operator $PHP - E - \Sigma(E)$ has mixed coefficient $+1/800$, despite the absence of an original X–Y interaction. The global scalar denominator retains the spectator energy. This does not contradict the v2.3 second-order onsite-resolvent coefficient, whose disjoint ordered edge terms vanish. It blocks only the automatic passage from that second-order fact to connectedness of the raw all-order global scalar Feshbach map.

6.2 Finite Ritz boundedness is not uniform SW smallness

Let

$$\Pi_M = \sum_{n=0}^{M} |n\rangle\langle n|, \qquad q = \frac{a+a^*}{\sqrt{2}}. \tag{6.4}$$

Each $\Pi_M q \Pi_M$ is a bounded finite matrix, but

$$\|\Pi_M q \Pi_M\| \geq \sqrt{M/2}. \tag{6.5}$$

For the correctly compressed quadratic bond,

$$\langle M,0|(q_x - q_y)^2|M,0\rangle = M+1. \tag{6.6}$$

Choose

$$B_M = \frac{1}{4}(\Pi_M \otimes \Pi_M)(q_x - q_y)^2(\Pi_M \otimes \Pi_M), \qquad \Gamma = 2. \tag{6.7}$$

Then

$$\frac{\|B_M\|}{\Gamma} \geq \frac{M+1}{8}, \tag{6.8}$$

which is at least 1/2 at $M = 3$ and at least 2 at $M = 15$. In the verifier's ASCII token form, the lower bound is `(M+1)/8`. Finite-cutoff boundedness therefore gives no cutoff-uniform ordinary operator-norm SW smallness at fixed Γ. This fixture does not reject relative-form smallness, graph norms, a connected QPS construction, or the resolvent-compatible Ritz-tail theorem in Section 5.

7 Devil's-advocate review

1. **DISMISSED only at fixed member – two Gibbs-L2 legs cannot control bounded observables:** cyclicity of the faithful standard vector and uniform boundedness give strong and adjoint-strong convergence on the whole Hilbert–Schmidt representation, so (3.8) follows. No observable-inserted Gibbs-L2 norm estimate is claimed.
2. **UPHELD as an overclaim – the $R^{-1/5}$ corridor is a thermodynamic strong-star rate:** changing the volume changes ρ, the standard vector, and the representation. The theorem fixes R_0 and sends only L to infinity.
3. **UPHELD – forward Cauchy alone gives an automorphism:** the v2.3 one-sided UHF fixture disproves that shortcut. Section 4 assumes both time signs over the full directed set.
4. **UPHELD as an overclaim – the conditional completion identifies Q3 dynamics or KMS phases:** it produces only an abstract C_0 group after the unproved Q3 hypothesis. It supplies no local generator domain and no state-limit theorem.
5. **DISMISSED – a factor two is missing in (5.3):** G_e is one skew off-diagonal block matrix, so $\|G_e\| = \|D_e\|$, not $2\|D_e\|$.
6. **DISMISSED – a global resolvent was silently declared local:** the proof retains one global R and localizes only its restriction to the exact high-support sector created by T_e.
7. **UPHELD as an overclaim – (5.7) proves all orders:** it is only the first generator and exact second-order low-block match. There is no third-order coefficient, summability radius, or uniform remainder.
8. **DISMISSED – the spectator fixture duplicates the extensive self-energy no-go:** the present failure is an exact mixed coefficient caused by a disconnected retained low-block energy in a scalar denominator, not the absence of decay information in one extensive norm.
9. **DISMISSED – finite Ritz boundedness implies uniform ordinary SW smallness:** (6.8) grows linearly with M at fixed gap.
10. **VALID with mitigation – verification pins prove the physical parents:** the three engines bind only the declared formulas, authorities, and scope. Their passing totals do not change the claim tier or close an open parent.
11. **DISMISSED – EXP-000817 contributes a fourth v2.4 theorem:** that row repairs historical authority links and wording only. It is cited as provenance, not as a theorem or result increment.

Independent adversarial review is invited especially on cyclicity of the right-commutant core, the use of both adjoint legs, passage of the group and inverse identities through a directed norm limit, locality of RT_e, the sharp off-diagonal norm, signs and the factor 1/2 in (5.7), the mixed Pauli coefficient in (6.3), the correctly compressed bond in (6.7), and every scope firewall.

8 Reproduction and exact verification pins

The R-167 v2.4 primary, non-importing independent, and integrated scripts are

1. codes/foundations/pre_a_cp1_st8_q3lock_local_measured_renyi_semiclassical_doublet_route_split.py;
2. codes/foundations/pre_a_cp1_st8_q3lock_local_measured_renyi_semiclassical_doublet_route_split_independent.py;
3. codes/foundations/pre_a_cp1_st8_q3lock_local_measured_renyi_semiclassical_doublet_route_split_verify.py.

From the repository root, run

```
Set-Location E:\Dev\TECT
$py = 'E:\Dev\TECT.venv\Scripts\python.exe'
$r167 = (
  'codes\foundations\pre_a_cp1_st8_q3lock_' +
  'local_measured_renyi_semiclassical_doublet_route_split'
)
& $py -X utf8 ($r167 + '.py')
& $py -X utf8 ($r167 + '_independent.py')
& $py -X utf8 ($r167 + '_verify.py')
```

The expected final totals are primary `PASS 343/343`, non-importing independent `PASS 198/198`, and integrated `PASS 341/341`. The exact raw verification-script SHA-256 pins are

```
R-167 primary:
62298845ace416840c85698d5aa6016099a1b532c1de9c3f6e88acd15912ad37
R-167 independent:
9d5f1d0bc2032841632361099a238fe9958e31c06d7bb13d6c6be33518c75a80
R-167 integrated:
87d3c52fbc9f1d468dd11a378670e32529ae454cd70c427965480aafd1634b82
```

The primary symbolic engine and the non-importing standard-library engine recompute the fixed-standard-form, all-shape completion, first-generator, spectator, and Ritz fixtures from labelled inputs. The integrated engine binds those computations to the exact exploration, result, gate, negative, strategy, generated-surface, stored-run, and checkpoint-lifecycle authorities. A failed or missing check invalidates the total; it is never converted into a weaker pass.

The document is R-167-only. R-168 v1.3 is historical and is not reissued; this section deliberately contains no R-168 pass count or hash. No per-lemma or intermediate PDF was issued.

9 Exact no-overclaim and dependency boundary

The common-alpha parent is **OPEN**. Fixed-finite point-strong-star convergence is not a common-representation thermodynamic limit. The conditional completion theorem assumes rather than proves bidirectional all-shape Q3 point-norm Cauchy control and identifies neither the local generator nor any KMS quotient.

The connected oscillator-elimination parent is **OPEN**. The first homological generator and second-order match provide no third-order coefficient, linked-cluster subtraction, convergent all-order interaction, uniform remainder, or cutoff-stable all-order QPS norm.

The broader infinite-dimensional rank-two parent is **OPEN**. No boundary/source-uniform parity-equivariant quasi-local elimination, phasewise oscillator intertwiner, or exact two-phase oscillator QPS transfer is proved.

The broken-sector spectral parent is **OPEN**. No oscillator temporal mass, oscillator-lattice GNS coercivity estimate, broken-sector GNS gap, or physical mass gap follows.

The Round-1 parent is **OPEN**. `EXP-000817` repairs provenance only, and `EXP-000818` changes neither the evidence-role freeze nor the prospective holdout state.

Nothing in this checkpoint proves regulator removal, continuum control, same-Hamiltonian physical-empty comparison, a below-empty sign, physical vacuum selection, C6 emergence, CP1, physical Sector A, or Pre-A closure. The registered `A5-SECTOR-A-SYNTHESIS` remains conditional at its stated scope, and the `C6-SPACETIME-SIGNATURE` claim card and tier are unchanged.

10 Result footer

```
Result ID:                 PA-CP1-ST8-Q3LOCK-SECOND-WEIGHTED-ENERGY-
                           MOMENT-AND-COMMON-ALPHA-CAUCHY-GATE-SPLIT.
Reusable result:           R-167 v2.4.
Theorem exploration:       EXP-000818.
Authority correction:      EXP-000817; linkage and wording only, no theorem or
                           reusable-result increment.
Precise statement:         Fixed finite faithful Gibbs standard-form strong-star
                           and bounded-observable conjugation cutoff removal;
                           conditional bidirectional all-shape point-norm Cauchy
                           completion to a unique C0 automorphism group; first
                           local parity-equivariant rank-two homological
                           generator with QPS/Ritz bounds and exact second-order
                           onsite-resolvent low-block match; two exact automatic-
                           promotion implication failures.
Claim context:             C6-SPACETIME-SIGNATURE; claim_bearing false.
Closed children:           Exactly the three R-167 identifiers in Section 2.
Negative results:          Exactly the two R-167 implication failures in
                           Section 2.
Open successor gates:      All-exhaustion common alpha; connected all-order
                           rank-two oscillator elimination/QPS and cutoff;
                           broader rank-two two-phase transfer; broken-sector
                           oscillator GNS gap; Round-1 evidence-role freeze.
Scope:                     T0 additive, claim-nonbearing analytic/exact result
                           in one fixed finite faithful Gibbs standard form, a
                           conditional common-C-star all-shape completion setup,
                           a first local homological-generator setup, and the
                           two finite exact counterfixture scopes.
Dependencies:              R-167 v2.3; EXP-000815; two-sided implementer legs;
                           faithfulness; actual bidirectional all-shape Cauchy
                           hypothesis; onsite high-support preservation; parity-
                           and resolvent-compatible Ritz-tail contract.
Evidence grade:            ANALYTIC + EXACT + primary + non-importing
                           independent + integrated; R-167-only final checkpoint
                           document; R-168 v1.3 historical and not reissued.
Reproduction command:      Run the three commands in Section 8 from repository
                           root, primary then independent then integrated.
Expected output:           PASS 343/343; PASS 198/198; PASS 341/341.
Primary raw SHA-256:       62298845ace416840c85698d5aa6016099a1b532c1de9c3f6e88acd15912ad37.
Independent raw SHA-256: 9d5f1d0bc2032841632361099a238fe9958e31c06d7bb13d6c6be33518c75a80.
Integrated raw SHA-256:  87d3c52fbc9f1d468dd11a378670e32529ae454cd70c427965480aafd1634b82.
Falsification gate:        Any standard-form cyclicity/adjoint-leg, fixed-member
                           rate, group/inverse limit, generator locality/sign/
                           factor, QPS/Ritz bound, spectator coefficient,
                           harmonic-cutoff lower bound, assertion count, hash,
                           provenance, or scope mismatch.
Tier before / after:       C6 T1 / C6 T1; no claim or tier change.
No-overclaim statement:    No moving-family strong-star rate, actual Q3 all-
                           shape common alpha, identified generator/KMS quotient,
```

	all-order oscillator elimination, phasewise oscillator QPS/GNS transfer, oscillator gap, Round-1 freeze, regulator removal, continuum, physical-empty sign, C6, CP1, physical Sector A, or Pre-A closure.
Next required action:	Prove both-time-sign full-directed-set Q3 point-norm Cauchy estimates and identify the local generator and phase-KMS quotients; independently derive and sum the connected all-order rank-two interaction with a uniform remainder and cutoff convergence, then prove the phasewise oscillator GNS coercivity estimate.
PDF policy:	One R-167-only gate-level checkpoint; no per-lemma or intermediate PDF. Preserve all earlier checkpoint pairs as historical evidence.
R-168 firewall:	R-168 v1.3 remains historical and is not reissued; no R-168 verification count or hash is part of this document.